

# NAMAN MANOCHA

AI ENGINEER • GENAI • AGENTIC SYSTEMS — LLMs, RAG & Applied ML

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## SUMMARY

AI/ML engineer with experience across classical ML, multi-agent LLM orchestration, and grounded RAG systems. Independently developed FundersAI, a research platform for Indian stocks and mutual funds, and evidence-first AI product work across Python, FastAPI, Next.js, PostgreSQL, Docker, and Kubernetes.

## TECHNICAL SKILLS

ML & AI: LLMs, LangGraph, Google ADK, RAG, Hybrid Search, Prompt Engineering, Hugging Face, Ollama, pgvector, ChromaDB, K-Means Clustering, Logistic Regression, Langfuse, MLflow, scikit-learn

Backend & Cloud: Python, FastAPI, REST APIs, SQL, PostgreSQL, Supabase, Redis, Docker, Kubernetes, GitHub Actions, Google Cloud Run, Vercel, Prefect, Git

Web & Data: Next.js, React, TypeScript, ETL Pipelines, Web Scraping, PDF Parsing, Data Validation, Data Visualization

## EXPERIENCE

### Artificial Intelligence Intern — Internmo (Remote)

Sept 2026 – Present

- Vantara Retail Solutions (client PRD: churn, CLV, segmentation, XAI): built leakage-safe pipeline on 1M+ retail transactions with 14 monthly point-in-time snapshots (53.7K rows), a time-based split, and Optuna tuning with forward-chaining CV.
- Delivered calibrated churn scoring (test ROC-AUC 0.76, recall 0.76), a LightGBM CLV model capturing 58% of value in the top decile, SHAP/LIME drivers, a recommendation engine, and FastAPI/PostgreSQL/Docker serving with PSI drift monitoring; 124 tests.
- FlowCast (Meridian Mobility PRD): building traffic forecasts for a 25-segment corridor (178K sensor rows): rule-based cleaning with cross-column repair, chronological splits with the test set touched once, and a from-scratch LSTM vs. classical ML.
- Northgate (quant research PRD): building leakage-free return forecasts for 10 US large caps over 9 years (4 ML and 4 deep-learning models vs. a random-walk baseline) with release-lag-aligned macro features, FinBERT sentiment, and a max-Sharpe optimizer.

### Technology Manager — PayGain Multiservices Pvt. Ltd.

Feb 2025 – Sept 2026

Kolkata, India

- Scoped and designed a company-wide mobile app (design stage), covering client/employee onboarding, document access, product activation, employee training, and payments — mobile counterpart to PayGain's core platform.
- Designed and deployed a lead-capture automation (Make) triggered on Google Form submissions, eliminating manual lead logging and improving sales outreach.
- Drove 30+ client conversions by building and managing the company website and digital discovery flows.
- Simplified complex investment/insurance products for 300+ seminar attendees via 20+ technical presentations and collateral — translating business requirements into deployed, client-facing systems.

## INDEPENDENT PROJECTS

### FundersAI — AI-Powered Research Platform for Indian Stocks & Mutual Funds

[fundersai.co.in](https://fundersai.co.in)

Python, FastAPI, Next.js, Supabase/pgvector, LangGraph, Groq, OpenAI, Cloudflare R2, GitHub Actions

Python, FastAPI, Next.js, Supabase/pgvector, LangGraph, Groq, OpenAI, Cloudflare R2, Kubernetes (K3s)

Solo-built, Apr–Aug 2026 — ~144K source LOC across 858 files, 443 commits, 179 test files — OpenAI Build Week | Live: [www.fundersai.co.in](https://www.fundersai.co.in) | Repo: [github.com/Naman6019/FundersAI](https://github.com/Naman6019/FundersAI)

- Citation-Grounded RAG: built a seven-stage LangGraph workflow with retrieval grading, query rewriting, citation-aware synthesis, lexical reranking, and explicit abstention when evidence is insufficient.
- Explainable ML:** engineered a fund-similarity engine using cosine similarity and k-means clustering over 13 standardized features, plus a logistic-regression triage model (MLflow) for parser-review prioritization.
- Multi-Agent Orchestration:** designed a custom multi-agent pipeline (Planner → Quant → Research → Synthesis agents) with shared state and real-time streamed status updates, where a Groq-hosted LLM router agent classifies query intent and directs execution across stock, mutual-fund, comparison, and market-news paths.
- Data-Quality Debugging: diagnosed and fixed scheme-name normalization defects, then added safeguards to preserve correct mutual-fund grouping and coverage.
- Resilient Data Engineering: automated official disclosure collection and normalization with source-specific parsers, data-readiness checks, and manual promotion gates.
- Cloud Storage & Billing:** built an S3-compatible storage layer (Cloudflare R2 via boto3/AWS SigV4, Python + TypeScript) for raw documents and cold archival, and shipped Razorpay subscription billing (webhook-verified, Free/Pro/Ultra) with Supabase Auth row-level security.

Python, FastAPI, LangGraph, Docker, Kubernetes (K3s), AWS EC2

- **Decoupled Architecture:** extracted report synthesis into a standalone FastAPI + LangGraph service, independently deployable from the main monolith so LLM-heavy generation scales without impacting the core app.
- **Kubernetes Service Operations:** provisioned and administered an AWS EC2 instance running K3s, hosting the report-synthesis service with two replicas behind a streaming API for real-time fund-comparison report generation.
- **Infrastructure & Secrets:** configured EC2 security groups/networking and managed credentials (Supabase, OpenAI) via Kubernetes Secrets rather than embedding them in the container image.

### CareFlow Intelligence — Agentic RAG & Fine-Tuning for Clinical Research

Python, FastAPI, Next.js, Supabase/pgvector, ChromaDB, Unsloth, Ollama | Solo-built, Aug 2026 — ~7,300 LOC — non-clinical, synthetic-data-only (MITRE Synthea)

- **Agentic Citation-Grounded RAG:** built a bounded-tool document agent with query reformulation and mandatory citation validation against PostgreSQL full-text indexes, backed by automated source ingestion.
- **RAFT Fine-Tuning:** designed a training-data pipeline with reference evidence, hard distractors, and abstention cases; fine-tuned Qwen2.5-1.5B via 4-bit QLoRA (rank 16, alpha 16), then quantized and served it locally through Ollama.
- **Local Model Serving:** quantized the fine-tuned model to GGUF and deployed it behind Ollama for zero-cost local inference, with an OpenRouter (Nemotron 3 Ultra) cloud fallback for the base agent.
- **Human-in-the-Loop Data Agent:** built a second agent for synthetic clinical data ingestion (10 patients, 479 encounters, 300 conditions, 5,368 observations, 296 medications across 18 FHIR R4 tables) with mandatory human-approval gates around autonomous data operations.
- **Full-Stack Delivery:** async FastAPI + SQLAlchemy + Alembic backend and Next.js 15 frontend, Dockerized end-to-end for local and cloud deployment.

### TalentOS — Autonomous Opportunity Intelligence Platform (All Things Agentic Hackathon)

Python, Google ADK, LangGraph, Next.js, Firestore, Google Cloud Run | Solo-built, Aug 2026 — ~21K source LOC across 116 files, 70 commits, 26 test files | Live: [all-things-agentic--allthingsage...](#) | Repo: [github.com/Naman6019/all-things-agentic](#)

- **Dual-Pipeline Agent System:** built Careers and Studio workflows with LangGraph StateGraphs and Google ADK agents on Gemini, aggregating ~2,900 postings/run across 4 ATS platforms and 4 aggregator feeds into reviewable outputs.
- **Cost-Disciplined Filtering & Evaluation:** applied deterministic title/seniority pre-filtering that eliminates 89.5% of postings before any model call, then used structured evaluator and drafter steps to prepare tailored application materials with persistent run state.
- **Guardrails & Observability:** shipped human-in-the-loop send gates, platform-compliance checks, Langfuse tracing, and a 373-test hermetic offline suite (~11s) for agent behavior.
- **Full-Stack Dashboard:** built a Next.js 15/React 19 dashboard with separate jobs and freelance views, printable tailored resumes, and an interactive pitch editor, deployed on Cloud Run.

### Question Answering on SQuAD — Model Fine-Tuning & Benchmarking

Python, PyTorch, scikit-learn, BERT, BiDAF, DistilBERT

- Compared BERT, BiDAF, and DistilBERT for extractive QA; fine-tuned each and built a standardized evaluation pipeline (accuracy, precision, recall, F1); published as a research paper.

### GitResume — Autonomous Resume & Portfolio Intelligence Agent

Python, Typer, Rich, Ollama, python-docx | ~4K LOC across 23 files, 28 commits | Repo: [github.com/Naman6019/git-resume-agent](#)

Self-built a multi-agent pipeline (Inspector → SchemaDiscoverer → Synthesizer → GroundingVerifier) that scans git history/AST, synthesizes quantified resume bullets via LLM, and adversarially verifies every metric against commit diffs before compiling to Word/PDF with native hyperlinks — published to PyPI as git-resume-agent; runs on every commit via a git hook.

## EDUCATION & ACHIEVEMENTS

### B.Tech, Computer Science Engineering (AI & ML Specialization) — VIT Amaravati

June 2021 – Sept 2025

CGPA: 8.2/10

- Published research paper: "Interactive Question Answering Model using Natural Language Processing" — International Journal of Modern Developments in Engineering and Science.
- Certifications: Azure Administrator Associate; AWS Certified Cloud Practitioner (Foundational).